Calvin Gacheru Mwangi

Data Scientist & AI Engineer — Agentic AI, RAG and Production ML

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SUMMARY

Final-year Data Science student at USIU-Africa who ships AI systems, not just notebooks. I build **LangGraph agents** and **RAG** pipelines on pgvector, train and explain ML models, and deploy them with **FastAPI**, **Docker** and CI/CD. I have delivered a live production website for a commercial client, and I am seeking graduate or junior roles in AI engineering, ML or data science.

SKILLS

Languages: Python (advanced), SQL, JavaScript, HTML/CSS, R, Bash

AI & LLMs: LangGraph, LangChain, RAG, tool calling, pgvector, Qdrant, Groq, DeepSeek, OpenAI API, Ollama

Machine Learning: scikit-learn, XGBoost, LightGBM, PyTorch, NLP, time-series forecasting, SHAP, Grad-CAM, cross-validation

Engineering & MLOps: FastAPI, Docker, GitHub Actions CI/CD, PostgreSQL, SQLAlchemy, Redis, Kubernetes, Linux, Git

Data & Security: pandas, NumPy, ETL, Power BI, Plotly, Streamlit | TLS/SSL, IAM/RBAC, KMS encryption, JWT auth

EXPERIENCE

Freelance Web Developer — BikBat Insurance Agency 2026
Nairobi, Kenya · live client engagement

- Delivered the client's business website as sole developer, from brief to production. I owned domain registration, DNS and hosting, and the site is live at bikbatinsurance.com.
- Structured service listings and trust-building content to convert visitors into insurance enquiries.

Data Science Intern — Prodigy InfoTech Sep 2025
Remote · completed with outstanding remarks

- Delivered four end-to-end projects, covering EDA, predictive modelling, customer churn and NLP sentiment analysis, each as a documented, reproducible repository.

PROJECTS

MyAnalyst — Autonomous Data Analysis Agent GitHub 2026
Python · LangGraph · FastAPI · PostgreSQL/pgvector · Docker · DeepSeek

- Built a LangGraph agent that profiles, cleans and forecasts uploaded datasets. It routes each question to a quantitative, document or hybrid track, and the hybrid track checks computed figures against document targets with citations.
- Ran LLM-generated pandas code in a network-isolated, non-root, resource-limited Docker sandbox, with automatic retries when a run fails.

MySaidi — AI Productivity Assistant GitHub 2026
Python · FastAPI · LangGraph · Groq (Llama 3.3 70B) · PostgreSQL · JWT · PWA

- Built a LangGraph agent that turns natural conversation into tool calls on tasks, calendar events and habits. Shipped it as an installable PWA with JWT authentication, deployed on Railway.

LexiAssist — Explainable Handwriting Screening GitHub 2026
Python · PyTorch · ResNet18 · Grad-CAM · Streamlit

- Fine-tuned ResNet18 against a CNN baseline, using writer-aware group 5-fold cross-validation and a decision threshold tuned for recall of at least 0.90. Served it in Streamlit with Grad-CAM heatmaps and a consent gate.

Uber Price Prediction — Production MLOps Pipeline GitHub 2026
Python · scikit-learn · FastAPI · Streamlit · Docker Compose · GitHub Actions

- Reached a mean **R^2 of 0.985** across 5-fold cross-validation on **93,000 rides** with a leakage-safe pipeline. Served it through FastAPI and Streamlit in multi-container Docker, with CI on every push.
- **Also:** RAG knowledge retrieval for technical support (LangChain, pgvector, Groq) | MamaGuard maternal-risk scoring (XGBoost, SHAP, SMS alerts) | diabetes screening (XGBoost + SMOTE, **88% recall**) | Twitter sentiment NLP (**93% F1**) | secure MinIO storage (TLS, RBAC, KMS; **100k** records) | all 20 projects

EDUCATION & CERTIFICATIONS

BSc Data Science & Analytics — USIU-Africa, Nairobi (final year) Expected 2027
Build AI Agents using LangGraph — Simplilearn Apr 2026
Data Science Internship, Certificate of Completion — Prodigy InfoTech Oct 2025

Languages: English (native), Swahili (native)